

User Guide Food Consumption Table (FCT) - Analysis using the 2015 Canadian Community Health Survey - Canada.ca

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1. Background

1. Purpose

Food intake data play an important role in regulations, policy-making, risk assessment and research activities in many organizations within the Department. The Bureau of Food

Surveillance and Science Integration (BFSSI) of the Food Directorate updated the food consumption table (FCT) using the data from the 2015 Canadian Community Health Survey, Nutrition (2015 CCHS Nutrition). The original FCT, based on the 2004 CCHS Nutrition data, was done in collaboration with the Bureau of Chemical Safety at the Food Directorate and the Existing Substances Risk Assessment Bureau of the Safe Environments Directorate at the Healthy Environments and Consumer Safety Branch. The ongoing aim of the FCT is to meet the increasing needs from evaluators and researchers to conduct exposure assessments using the most recent Canadian food intake data and to ensure consistent use of these data within Health Canada.

2. What is the Food Consumption Table (FCT)?

The FCT, available in a Microsoft Excel spreadsheet, provides estimates of daily intakes of foods (means and percentiles) obtained using data from the 2015 CCHS Nutrition. Estimates were generated at three different food grouping levels based on the food group list from the Bureau of Nutritional Sciences (BNS). These estimates are presented for two consumption populations: “all persons” and “eaters only”. Each estimate is further broken down by various age-sex categories. For both consumption populations, estimates are available in two different units: 1) grams per person per day and 2) grams per kilogram of body weight per day. The *Read Me*, *Food Group List* and *Body Weights* tabs of the FCT provide detailed information to facilitate its use.

3. What has changed between the original Food Consumption Table and this update?

Some features have been changed from the previous version of the Food Consumption Table that was published in 2014. Here is an overview of these changes:

1. Age groups:

- In the design of the 2015 CCHS Nutrition, sampling of participants starts at age 1. In 2004, data was collected from age 0. The age groups “0-5 months”, “6-11 months”, “1 year” and “2-3 years” that were present in the previous FCT have now been replaced by a single age group of “1-3 years”.

2. Food groups and subgroups:

- A new food group (54) was created for supplemented bars, shakes and meal replacements.

- Twelve new food subgroups were created: 7C Granola bar, 10J Plant based-beverage, 10K Goat and sheep milk, 46E Energy drink, 46F Vitamin water, 46G Sports drink, 54A Energy bar, 54B Protein bar and shake, 54C Meal replacements, 231H Infant formula prepared, 231I Liquid meal replacement, and 231J Protein powder.
- Also the following 4 food subgroups have been added to the table: Soft Drink (46A&B regular and diet combined), Water from all sources (51C and water from recipes), wheat flour and corn flour separately.

3. Calculation of imputed body weights:

- The approach to imputation of missing body weight for those aged 2+ has been updated. The missing body weights are imputed using a regression model with demographic, geographic, nutritional and health predictors features within each age & sex group.
- Imputation method for missing body weights of children aged 1 and 2 remains unchanged.

4. Baby food products:

- As mentioned above, the distributions for age groups “0-5 months” & “6-11 months” are no longer available. Intakes of baby food products are available for the age group “1 – 3” and for other age groups.

4. What is the 2015 Canadian Community Health Survey with a Nutrition Focus?

The Canadian Community Health Survey consists of a series of cross-sectional surveys that were initiated in the year 2000, with a main objective of providing timely information on health determinants, health status and health system utilization. It stems from a partnership among Health Canada, the Canadian Institute for Health Information, and Statistics Canada. The CCHS is comprised of two types of surveys: an annual component dedicated to general health and a more focused survey on specific health topics. The focused surveys are designed to provide reliable estimates at the national and provincial levels. In 2004 and 2015, the theme of the CCHS focused survey was nutrition.

The main objective of these nutrition focused surveys was to provide reliable, timely information about dietary intake, nutritional well-being, and their key determinants. This

was to inform and guide programs, policies and activities of federal and provincial governments.

The target population of the 2015 CCHS Nutrition included individuals aged 1 year and older living in private dwellings in the 10 Canadian provinces. A total sample of 19,670 was selected through a complex sampling plan to ensure that the sample represents the Canadian population with regard to age, sex, geography and socioeconomic status. The target population did not include individuals who were full-time members of the Canadian Forces or who lived in the Territories, on First Nation Reserves, in prisons or care facilities, or in some remote areas.

More detailed information on this survey can be found in the document “Reference Guide to Understanding and Using the Data: 2015 Canadian Community Health Survey – Nutrition” developed by Health Canada.^{[Footnote1](#)}

2. Methodology Used for this Project

1. Data Source

In the 2015 CCHS Nutrition, detailed food intake information was obtained through 24-hour dietary recalls at the time of interview. Permission was asked to all respondents to allow sharing the information collected during the survey with partners such as Health Canada. Only those persons who agreed to share their information with Health Canada are contained in the Share file. The total number of respondents in the Share file is 19,670 individuals.

2. Subjects

The analysis included all respondents (including pregnant and lactating women) in the Share file who reported one or more food items consumed in their 24-hour recalls.

The subjects were classified into different age-sex groups based on the Dietary Reference Intakes (DRIs) age-sex groups. The FCT working group recommended the inclusion of subtotals for ages 1-18 (both sexes combined) and 19+ years (sexes combined or separated). All children aged less than 4 were combined into the 1-3 age group. A total of 25 different age-sex groups are tabulated.

3. Food Groups

Estimates were generated for three different levels of food grouping – main food group, food group and food sub-group based on the BNS food group list. Ten main food groups (Grain products, Dairy products, Fats, Meats, Meat alternatives, Vegetables, Fruits, Beverages, Babyfood products and Miscellaneous) were defined by the FCT working group and the

intakes are presented on ten separate tabs on the FCT (see the tab of Food Group List for a full listing in the Excel file).

A new food group (54) was created for supplemented bars, shakes and meal replacements.

Twelve new food subgroups were created: 7C Granola bar, 10J Plant based-beverage, 10K Goat and sheep milk, 46E Energy drink, 46F Vitamin water, 46G Sports drink, 54A Energy bar, 54B Protein bar and shake, 54C Meal replacements, 231H Infant formula prepared, 231I Liquid meal replacement, and 231J Protein powder.

Also 4 food subgroups have been added to the table: Soft Drink (46A&B regular and diet combined), Water from all sources (51C and water from recipes), wheat flour and corn flour separately.

Notes about these and other updates can be found on the column of Food Group Name (column A) of the FCT where appropriate.

4. Estimates

Arithmetic mean and percentiles [i.e. 50th (P50), 90th (P90) and 95th (P95)] of daily intakes of foods were calculated for each of the food groups using data from the first 24-hour dietary recall. Estimates of daily intakes are presented by the age-sex groups agreed upon by the FCT working group. This was done for “all persons” and for “eaters only”. Under each consumption population, estimates are expressed as both grams per person per day (g/person) and grams per kilogram of body weight per day (g/kg body weight/day). Standard errors (SEs) were also calculated for each mean and percentile. All estimates were adjusted according to the instructions from Statistics Canada by using the sampling weights provided in the 2015 CCHS Nutrition data files in order to reflect the Canadian population. [Footnote2](#)

5. Body Weight

Body weights were used for calculating intakes in grams per kilogram of body weight per day for the FCT. In the 2015 CCHS Nutrition, body weight data (measured or self-reported) are available for respondents aged two years or older. Overall, 74.5% (14,661) of all respondents included in the analysis agreed to have their body weight measured and 19.0% (3,742) had a self-reported value. For respondents whose body weight values were missing (6.45%), the FCT working group decided that it was important not to eliminate their 24 hour recall data.

Imputation was performed to assign body weight values for missing ones. For those two years or older values were imputed as follows:

1. Determine the covariates which have significant predictive power toward measured body weights in each DRI group;
2. Impute the missing body weights with a regression model that utilized the previously selected covariates in each DRI groups. The standard errors of the regression parameters were also incorporated to retain the variability of measured body weight observations.

For those under the age of two, any missing body weights were imputed with the values provided by the US Centre of Disease Control (CDC) using data from the U.S. National Health and Nutrition Examination Surveys (NHANES), 1999-2010 (available upon request to the BFSSI).

6. Data Release

The data presented in the FCT follows the data release guidelines from Statistics Canada.

1. Estimates with a coefficient of variation (CV) between 16.6% and 33.3% should be used with caution. These are identified with the letter E next to the estimates.
2. The estimate cannot be reported when any one of the following situations is present. All estimates that are not reported due to situations listed below are identified by the letter F or a decimal point (.).
 - a. the number of eaters is less than 30
 - b. the CV is greater than 33.3%
 - c. the number of eaters is less than 100 for the 90th percentiles
 - d. the number of eaters is less than 200 for the 95th percentiles

7. Software for Calculations

Statistical Analysis System Enterprise Guide (SAS EG) 4.2 was used to calculate means and percentiles and their corresponding standard errors.

3. Content of the FCT

A total of 19,670 respondents were included in the analysis for all food groups.

The FCT file contains 13 tabs: 3 of them are “Helper Tabs” and the remaining 10 constitute the daily food intake tables for each of the ten main food groups. The Helper Tabs are set up to help users better understand and use the FCT data. They contain: a Read Me tab, a Food Group List tab and a Body Weights tab.

Estimates were generated for 10 main food groups, 48 food groups and 155 food sub-groups. Each tab presents the values for the overall main food group first and followed by each designated food group and food sub-group. For each food entry, estimates are presented for each of the 25 age-sex groups. These groups include: 1-3 years, 4-8 years, 9-13 years, 14-18 years, 19-30 years, 31-50 years, 51-70 years, 71+ years, 1-18 years, 19+ years and all ages (sexes combined), and males and females separately for ages 9-13 years, 14-18 years, 19-30 years, 31-50 years, 51-70 years, 71+ years and 19+ years.

Estimates are also presented for the two consumption populations - all persons and eaters only. Under each consumption population, estimates are expressed by grams per person per day and grams per kilogram of body weight per day, respectively. Specific details (including food regrouping, and food item exclusion) are noted on the first columns under the corresponding food group name and code.

The Read Me tab was designed as a quick overview of the FCT. It includes all key information about the table including the objective of the FCT, notes, comments, legends as well as how to cite data from the FCT and where to find further help.

The Food Group List tab contains a full listing of all food groups at the three food grouping levels (main food groups, food groups and food sub-groups) based on the BNS food group list. It also provides a description of the content of each food group as well as the notes for specific sub-groups, to allow users to rapidly determine where to look for a given food. The Body Weights tab presents the mean and median body weights of respondents by the 25 age-sex groups included in this analysis. It is included as a benchmark when needing average body weights for all age-sex groups.

4. How to Cite Data from the FCT

For published articles, studies, and reports using data extracted from the FCT, suggested citation for the FCT is as follows:

Health Canada (2018). Food Consumption Table derived from Statistics Canada's 2015 Canadian Community Health Survey, Nutrition, Share file. Ottawa.

5. Limitations in Table Use

The FCT provides estimates of daily intakes of foods at the predetermined food grouping levels only. Since it is designed for generic use in food and substance exposure assessments, there are some limitations that users should keep in mind. They are as follows:

1. All FCT estimates are daily intake estimates calculated from the first 24-hour dietary recall and reflect food intake amounts on a given day. They are not usual intake

estimates, which are calculated from the two dietary recalls using a more complex procedure. Therefore, the estimates from the FCT *should not* be interpreted as usual amount consumed where "usual" represents the average daily consumption if we could measure intake over a sufficiently long time period.

2. All estimates of daily intakes for food groups were generated by combining foods consumed as a food on its own and ingredient in recipes. For example, intakes of eggs on the FCT are the total consumption amount of eggs coming from hard-boiled egg (food) and eggs used in recipes such as quiches, homemade cakes, etc. The absolute daily intakes of some ingredients such as eggs, sugar, flour, etc. may be underestimated due to the inability to decompose some commercial food products.
3. Intake data are available for the 15 Dietary Reference Intakes (DRIs) age-sex groups and limited customized age-sex groups. Values from different age-sex groups presented in the FCT *cannot* be combined. For example, it is not possible and not acceptable to obtain an intake value for the 1-8 years old group by combining intake values from the two age groups of 1-3 years and 4-8 years old. For this purpose, the combined groupings "1-18" and "19+" are available, but if more specific groupings are required, please contact the BFSSI for support.
4. Intakes from different cooking methods such as grilling, barbecuing and oven-roasted cannot be separated
5. Intake data for foods that are not commonly consumed might be absent for some age-sex groups on the FCT to comply with the terms and conditions of the data release guidelines of Statistics Canada. Estimates might be available by the collapsing of age groups. If these estimates are required, please contact the BFSSI for support.
6. The 2015 CCHS Nutrition was conducted in the ten Canadian provinces. Food intake data for Aboriginal Canadians living on-reserve are not available on the FCT.
7. Comparison between the intakes of foods found in the original FCT and the update cannot be done directly due to methodological changes between the two surveys

Any request beyond what is present on the FCT can be made by emailing to: hc.nutrition.surveillance-nutritionnelle.sc@canada.ca with the subject line including "FCT request".

6. References

Footnote 1

Health Canada (2018). 2015 Canadian Community Health Survey - Nutrition - Reference Guide to Understanding and Using the Data. Available at: https://www.canada.ca/content/dam/hc-sc/documents/services/food-nutrition/food-nutrition-surveillance/ReferenceGuide2015CCHS-Nutr_Eng_Final_06192017.pdf.

[Return to footnote1referrer](#)

Footnote 1

Statistics Canada (2017). 2015 Canadian Community Health Survey (CCHS) – Nutrition User Guide. Available by request at: statcan.hd-ds.statcan@canada.ca